RO Antiscalant Projection Engine

Technical review & improvement comments for the Chembond Flux development team

Prepared for: Rahil (tool author) → Chembond Chemicals web development team
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Source reviewed: GitHub repository **rahils20/Reliance-MED-Performance** (branch head 030a52a): projection_engine.py, streamlit_app.py, med_suite.py, calculator_tab.py, run_app.py, requirements.txt, .github/workflows/build-exe.yml

1. Scope and status

This review covers the original Python/Streamlit tool that Chembond Flux is based on. The calculation engine was read line by line, and its key functions were run on benchmark waters to confirm each finding with real numbers rather than by reading the code alone. Everything marked “verified” below was reproduced by running the code.

The Chembond Flux website itself was not reviewed in this pass. The review ran in a cloud session that could not reach the browser on the author's PC, so no observations about the live site are included. Section 2 gives the web team an acceptance-test protocol and benchmark cases with expected outputs, so they (or a follow-up review) can check the port against the original engine.

Headline findings

- **Sulfate and fluoride saturation is overstated by about 3×** at typical RO concentrate salinity. The activity correction applied to the solubility products uses the wrong charge factor (verified: about ×3 at I ≈ 0.12 M). $CaSO_4$, $BaSO_4$, $SrSO_4$ and CaF_2 ratios, and therefore product selection and dose, are biased.
- **Zero calcium or zero bicarbonate switches off every scaling check.** A softened feed (Ca = 0) containing barium, strontium and sulfate reports zero risk for all salts, then tells the user that no catalogue product works (verified, case E). This matters for HERO-type plants, which run on softened feed.
- **Undersaturated calcium carbonate is treated as a failure.** The selector requires the treated LSI to be between −0.5 and +0.5. If the concentrate LSI is below −0.5 (soft or acidified water), every product fails and the UI shows “Standard Catalog Limits Exceeded – custom formulation required” (verified, case D).
- **The antiscalant model is an uncalibrated heuristic with no upper limit.** It lowers the thermodynamic SI itself. In one test, 7 ppm of a homopolymer-only product “controlled” gypsum at 26× saturation. No real antiscalant can do that, and supplier limits are typically about 2–4×.
- **The customer-facing report makes claims the model does not support.** The executive summary always says “severe scaling risk… permanent membrane damage is imminent”, even for benign water. The CIP-frequency and ROI figures are hard-coded, and the $CaCO_3$ row of the report table always prints 1.000.
- **Security:** the admin override code and the master password are hard-coded in the source. If Flux runs the engine in browser JavaScript, the proprietary product compositions and k-factors are visible to any customer.

Severity scale

Critical	Wrong answer or misleading customer output in common cases; fix before external use.
High	Material accuracy error or failure in a realistic scenario; data loss; security exposure.
Medium	Limited accuracy impact, edge-case defect, or misleading labelling.
Low	Code quality, maintainability or cosmetic issue.

2. Chembond Flux – acceptance test protocol for the web port

The web team should enter the five cases below into Chembond Flux and compare the outputs. These values were produced by the original engine using the same pipeline as the Streamlit UI. Two defaults matter: ion balance is auto-corrected (Na/Cl added) whenever the imbalance is above 5%, and auto-acid is on. **The expected values include the original engine's known errors (Section 3).** A mismatch therefore means one of two things: a porting error, or a deliberate fix. Every deliberate fix should be documented.

Common inputs unless stated: Ca 150, Mg 50, Na 300, K 10, NH_4 0, Ba 0.05, Sr 1.2, Fe 0.02, Al 0.01, HCO_3 250, Cl 400, SO_4 200, F 0.5, NO_3 5, PO_4 0, CO_3 0, SiO_2 15, CO_2 5 mg/L (the app's default water); 99% rejection; BWRO.

Case	Inputs (changes from default)	Treated pH / H_2SO_4 ppm	Conc. pH / LSI	I (M)	$CaSO_4$ / $BaSO_4$ / $SrSO_4$ / CaF_2 ratio	Expected recommendations (ppm)
A	Default water, pH 7.5, 25 °C, 75% recovery. Balance error 11.85% → Cl raised to 587.3	7.50 / 0.00	8.102 / 2.065	0.1313	1.684 / 35.84 / 0.457 / 2.372	R 4001 7.5 · R 428 I 7.0 · R 6196 4.0 · R 6863 5.0 · R 170 5.5 · R 3687 5.0
B	Ca 200, HCO_3 400, SO_4 300, Ba 0.1, Sr 2.0, SiO_2 30; pH 7.8, 30 °C, 80%	7.30 / 21.56	7.999 / 2.472	0.1893	4.888 / 156.04 / 1.657 / 5.757	R 170 7.0 only
C	Soft: Ca 10, Mg 3, HCO_3 30, SO_4 20, Cl 60, Na 40, Ba/Sr/F 0; pH 7.0, 75%	7.00 / 0.00	7.602 / −0.207	0.0132	0.022 / 0 / 0 / 0	All 10 products at 1.0 ppm. The report still says “severe scaling risk”
D	Ca 100, HCO_3 8, SO_4 1500, Na 300, Cl 200, Mg 20; feed pH 6.0, manual acid, 75%	6.00 / 0.00	6.602 / −1.176	0.2242	6.839 / 218.33 / 2.783 / 1.425	None → “Catalog limits exceeded” (bug B1)
E	Softened: Ca 0, Mg 0.5, Ba 0.15, Sr 3.0, HCO_3 200, SO_4 400; pH 7.5, 75%	7.50 / 0.00	8.102 / −5.000	0.0000	0 / 0 / 0 / 0	None (bug B2: all indices zeroed)

Website checks to perform

- **Numerical parity:** compare cases A–E to 3 decimal places, including the acid dose, ionic strength, every saturation ratio, the product list and order, and the required doses. Also check the 1.0–8.0 ppm search step (0.5 ppm).
- **Ion balance behaviour:** confirm whether Flux silently changes Na/Cl. The original changes the calculation before the user presses “Apply Adjustments”, and the app's own default water is 11.9% out of balance. The adjusted analysis should be shown clearly and printed in the report.
- **Where the engine runs:** if calculations run in browser JavaScript, the product compositions (active %), k-factors and the admin override code are exposed to anyone through DevTools. Keep the engine server-side behind an API.
- **Bug carry-over:** retest every Critical/High item in Section 3 on the website, especially B1, B2, B3 and M1, and confirm whether each was fixed or ported.
- **Units and labels:** state product dose vs active dose, feed-basis ppm, the H_2SO_4 strength (98%), and that L/h assumes SG 1.15 (MED module uses 1.20 for R 3687).
- **Input validation:** negative or blank values, recovery ≥ 95%, temperatures outside 5–45 °C, pH outside 4–10, Ca = 0 or HCO_3 = 0, and very high TDS (SWRO).
- **Report:** the downloaded report must contain the full water analysis, the operating conditions, the acid dose, the chemical flow rate, assumptions, model version and a disclaimer. Remove the boilerplate claims (B3, B4).
- **General web QA:** mobile layout, browser compatibility, page load time, error messages instead of blank panels, accessibility (contrast, keyboard use), audit log of who generated which proposal, and the ability to save and reload projects.

3. Calculation engine – accuracy findings (projection_engine.py)

What is correct: the carbonate constants (Harned–Scholes pK_2 = 10.330, Plummer–Busenberg calcite pK_s = 8.480, Harned–Davis pK_1 = 6.351 at 25 °C), the Langelier pH_s expression with Davies activity coefficients, the H_2SO_4 stoichiometry (98.08 per 2×61.02, ÷0.98) and the chemical feed-rate arithmetic were all checked and are sound. The issues below concern the other salts, the antiscalant model and the system mass balance.

3.1 Thermodynamics & water chemistry

ID	Severity	Finding, location & evidence	Recommendation
M1	**Critical**	**Wrong charge factor in the Ksp activity correction.** get_pKsp_apparent() (lines 136–152) uses z = 4 for $CaSO_4$/$BaSO_4$/$SrSO_4$/$CaSiO_3$ and z = 2 for CaF_2. Debye–Hückel requires Σz_i^2 = 4+4 = 8 for a 2:2 salt and 4+2×1 = 6 for CaF_2. Verified: at the default concentrate (I ≈ 0.12–0.13 M) the apparent Ksp is understated about 3× (×2.99–3.09), so every sulfate/fluoride ratio is about 3× too high. Example: case A $CaSO_4$ shows 1.68 (“High”) where the corrected value is about 0.55 (undersaturated).	Use the same Davies γ already computed for LSI: $Ksp' = Ksp / (\gamma_{cat}\cdot\gamma_{an}^{n})$. Use one activity model throughout. Add unit tests against PHREEQC for 3–5 reference waters.
M2	**High**	**No ion pairing.** Free-ion activities ignore the $CaSO_4^\circ$, $MgSO_4^\circ$, $NaSO_4^-$, $CaHCO_3^+$ and $CaCO_3^\circ$ complexes. In sulfate-rich or high-TDS concentrates this overstates sulfate scaling (by tens of percent at high SO_4) and also biases LSI.	Use a proper speciation engine: PHREEQC (phreeqpy/IPhreeqc, with phreeqc.dat or pitzer.dat for SWRO) on the server. At minimum, add $CaSO_4^\circ$ and $MgSO_4^\circ$ pairing.
M3	**High**	**No ionic strength or other index when Ca = 0 or HCO_3 = 0.** Lines 108–118 return LSI = −5 and zero for every sulfate, fluoride and silica index. Verified (case E): softened feed with Ba 0.15, Sr 3.0, SO_4 400 returns I = 0 and $BaSO_4$ = $SrSO_4$ = 0. The same happens with fully dealkalised feed.	Only skip the LSI/SDSI calculation when Ca or HCO_3 is zero; compute all other salts normally. (Also see B2 for the selector impact.)
M4	**Medium**	**Silica limit has no pH or temperature basis beyond a linear fit.** The limit is fixed at 125 mg/L at ≤ 25 °C (lines 166–171) with no pH correction. Verified: the ratio is identical at concentrate pH 6.0, 7.0, 8.0 and 8.5. Amorphous silica solubility rises sharply above about pH 8.5, and vendor programs apply a pH correction factor. The same limit is coded four times (lines 166, 299, 947, 1145).	Implement the standard temperature + pH correction for amorphous silica (with Mg/Al co-precipitation warnings). Define the limit once in a shared function.
M5	**Medium**	**Metal silicate model is not referenced.** The SiO_3^{2-} fraction uses pKa_2 = 11.8 (literature for H_4SiO_4 is about 13). pKsp($CaSiO_3$) = 8.0, and Mg = 1.5× / Fe = 0.05× of the Ca value, have no source. A hard cut-off at pH 8.0 makes results jump. The report prints $CaSiO_3$/$MgSiO_3$ rows as if they were validated.	Remove these indices from customer outputs until validated, or replace them with an established magnesium-silicate guideline (Mg × SiO_2 vs pH).
M6	**Medium**	**The carbonate system is over-specified.** Feed pH, HCO_3 and CO_2 are entered independently and never reconciled; the CO_2 input does not affect any result. CO_2 released by acid (≈0.72 mg per mg HCO_3 destroyed) is not added to the treated feed. Concentrate pH = feed pH + $\log_{10}$CF (lines 496, 523) assumes 100% CO_2 passage and ignores CO_3^{2-}. This approximation is acceptable, but it is undocumented.	Calculate CO_2 from pH and alkalinity, and warn when the entered CO_2 disagrees. Carry the acid-generated CO_2. Document the concentrate-pH method (e.g. ASTM D4582) in the UI and report.
M7	**Medium**	**Mass balance is not consistent.** The concentrate uses C_f×CF and ignores salt passage (exact value: $C_f(1-Y\cdot SP)/(1-Y)$; ≈1% difference at 99% rejection). Permeate is computed from the untreated feed rather than the acid-treated feed, and from the feed concentration rather than the average feed–concentrate concentration. **Membrane type has no effect on any calculation**; it only changes the recommended-recovery caption.	Use the exact mass balance with the treated feed. Either remove the membrane-type selector or tie it to rejection, recovery limits and the SWRO activity model.
M8	**Medium**	**Constants fixed at 25 °C.** Sulfate and fluoride pKsp values ignore temperature, although the input allows 1–50 °C ($BaSO_4$ and CaF_2 solubility change noticeably). pKsp sources are not cited ($SrSO_4$ 6.49 vs ≈6.63 and CaF_2 10.40 vs ≈10.5–10.6 in common databases).	Add temperature-dependent log K (van 't Hoff or analytic expressions from phreeqc.dat). Cite the source of each constant in the code and the report.

ID	Severity	Finding, location & evidence	Recommendation
M9	**Low**	**SDSI is not Stiff-Davis.** It is the LSI with a different activity term (line 133), not the empirical Stiff–Davis K. The selector also forces both LSI and SDSI into ±0.5, although SDSI is meant for TDS above about 10,000 mg/L and LSI for brackish water.	Pick the index by TDS (LSI for brackish, S&DSI for SWRO), or show both with clear applicability notes.
M10	**Low**	**TDS and balance details.** Calculated TDS adds dissolved CO_2 (a gas) to TDS. The Fe equivalent weight assumes Fe^{2+}. “Fe/Al SI” is just mg/L ÷ 0.05, a feed guideline that is then applied to the concentrate and plotted on an SI axis. The app's own default water is 11.9% out of balance, so auto-balance runs on first load.	Exclude CO_2 from TDS. Label Fe/Al as a feed-guideline ratio and evaluate it on the feed. Ship a balanced default water.

3.2 Antiscalant performance model & product selection

ID	Severity	Finding, location & evidence	Recommendation
M11	**Critical**	**The antiscalant model changes the thermodynamic SI and is not calibrated.** calculate_effective_scaling() (lines 241–279) sets SI_{eff} = SI·exp(−Σ dose·active%·k / √SI). Antiscalants do not change saturation. They extend induction time, which raises the saturation the system can tolerate. The k-factors (lines 43–53) have no data source. **There is no upper limit:** verified that R 824 (homopolymer only) at 7 ppm “controls” $CaSO_4$ at 26× saturation (about 9× even after the M1 fix). Supplier limits are typically about 2–4× for gypsum and LSI about 1.8–2.8.	Reformulate the model as product maximum allowable limits vs dose (LSI_{max}, $CaSO_4$×, $BaSO_4$×, $SrSO_4$×, CaF_2×, SiO_2 mg/L), derived from Chembond induction-time/jar tests and field history, and compare them with the actual concentrate values. calculator_tab.py already produces such a limits matrix: connect it, then validate it. Keep reporting the real SI.
M12	**High**	**$BaSO_4$/$SrSO_4$ can never fail the selection.** A salt only fails if its “excess mass” is above 0.5 ppm (line 334). The maximum possible $BaSO_4$ in the default concentrate is 0.34 mg/L, so barium can never govern, even at 36× saturation (case A) or 156× (case B). Barium sulfate is one of the most damaging and least reversible RO scales.	Judge sparingly soluble salts (Ba, Sr, CaF_2) on saturation against the product limit, not on mass. A mass threshold is only meaningful for $CaCO_3$/$CaSO_4$.
M13	**Medium**	**The excess-mass calculation is arbitrary.** $CaCO_3$ mass is multiplied by 0.05 with no stated basis (lines 293, 942, 1139). The (1−1/ratio) precipitable fraction is only approximate. For silica, the excess above the limit is multiplied by (1−1/ratio) again, which double-counts. The CaF_2 exemption up to SI 1.5 (line 335) is undocumented. This function is copy-pasted three times.	Calculate precipitable mass by solving the equilibrium (mass-balanced IAP = Ksp′) or with PHREEQC. Keep one shared implementation.
M14	**Medium**	**The “economic” order ignores dose and price.** economic_sort_order (line 283) is a fixed list. R 4001 is ranked first although it needs 7.5 ppm vs 4.0 ppm for R 6196 (case A).	Rank by cost per m³ = dose × price (and by product availability). Show the cost per m³ and per year in the table.
M15	**Medium**	**Auto-acid only looks at LSI.** Lines 486–505 target concentrate LSI ≤ 2.5, a fixed value that is not tied to the chosen antiscalant. H_2SO_4 adds sulfate, which raises $CaSO_4$/$BaSO_4$. Verified: acid to pH 6 raised the $CaSO_4$ ratio from 1.74 to 2.84. HCl is not offered. If LSI ≤ 2.5 is not reached by pH 4, no acid is applied and no warning is shown.	Optimise acid against all the limits together. Offer HCl. Warn when the target is not reachable. Tie the LSI target to the selected product's limit.
M16	**Low**	**Dose basis is ambiguous.** The Project X slider is labelled “Active Product Dose” but is multiplied by active % (line 264), so it is really the product dose. The UI never says whether the dose is on a feed or concentrate basis.	Label it “Product dose, mg/L on feed flow”, and optionally show the active dose and the concentrate-side concentration.

4. Engine logic defects & customer report (projection_engine.py)

ID	Severity	Finding, location & evidence	Recommendation
B1	**Critical**	**Undersaturated LSI is treated as a failure.** Lines 323–324 require $-0.5 \le LSI_{eff} \le 0.5$. An antiscalant cannot raise LSI, so a concentrate with LSI < −0.5 fails for every product. The user then sees “Standard Catalog Limits Exceeded... Custom formulation required” and an admin-code prompt. Verified: case D (LSI −1.18, $CaSO_4$ 6.8×) gets zero products, while the same engine accepts 26× $CaSO_4$ when LSI is positive.	Apply an upper limit only (LSI ≤ product limit). Treat negative LSI as “no $CaCO_3$ risk” and, if very negative, show a corrosivity note.
B2	**High**	**A softened or dealkalised feed causes a false “no product works”.** Because of M3, all indices are zero and LSI = −5, so B1 rejects every product (case E). A HERO plant (softener + HRU upstream) cannot be simulated.	Fix M3 and B1, then add regression test case E.
B3	**Critical**	**The executive summary is hard-coded.** Lines 907–912 always state “severe scaling risk... rapid loss of flux and permanent membrane damage is imminent” and that the product will “completely suppress” the salts. This is printed even for case C (LSI −0.21, every product passes at 1 ppm). It is a commercial and legal liability in a customer proposal.	Generate the text from the results (risk class by salt). Use measured language (“projected”, “based on the supplied analysis”). Add assumptions and a disclaimer.
B4	**High**	**CIP and ROI figures are invented.** In lines 1039–1051 the “b_max_mass_ppm” variable adds up saturation ratios, not masses. Using 10^{LSI} means any LSI above about 1.7 gives “CIP every 1–2 weeks”. The treated case is hard-coded to “CIP every 3–6 months, membrane life optimal, energy stable” whatever the dose or water.	Remove this section, or base it on site history or a documented fouling model. Never show it as a projection without a basis.
B5	**High**	**The $CaCO_3$ row of the report table is always 1.000.** Line 968 looks up “Ratio_CaCO3”, which does not exist, so the default of 1.0 is printed as the baseline.	Show the LSI (or 10^{LSI}) for $CaCO_3$ consistently, or drop the duplicate row.
B6	**High**	**Report content is incomplete.** The .docx omits the water analysis (raw and balanced), temperature, pH, flow, CF, acid dose, chemical flow (L/h), the product limits used, and the software/model version. Charts are silently omitted when kaleido cannot render (kaleido ≥ 1.0 needs Chrome). Temporary PNG files are never deleted (delete=False, line 1014).	Add an inputs annex and an assumptions/version block. Use BytesIO for images, or render charts with matplotlib (already a dependency).
B7	**Medium**	**Errors are hidden.** calculate_scaling_indices() catches every exception and returns None (lines 238–239), and the UI then shows only “ensure Ca and HCO_3 > 0”. Results depend on `'treated_conc_data' in locals()` (lines 800, 890).	Validate inputs explicitly, show the actual error, and pass results as return values or a dataclass.
B8	**Medium**	**Custom blend issues.** Percentages are not checked to sum to ≤ 100% (lines 836–850). The success message is lost because st.rerun() runs straight after it. The custom blend stays in the session and pollutes later product lists.	Validate the blend total, persist the message across the rerun, and clear the blend on reset.
B9	**Low**	**Other UI issues.** The finalise-dose input allows up to 10 ppm, but the search stops at 8. The print CSS relies on auto-generated .st-emotion-cache class names that change between Streamlit versions. st.set_page_config() runs when the module is imported (line 10), before the hub's own call: older Streamlit raises an error, and newer versions depend on call order.	Align the limits, move the set_page_config call under __main__, and pin the Streamlit version.

5. Security, data integrity & RO monitoring hub (streamlit_app.py, med_suite.py)

ID	Severity	Finding, location & evidence	Recommendation
S1	**Critical**	**Hard-coded secrets.** The admin override “KEMPRO2026” is in projection_engine.py line 826, and the master password “12345678” appears 3× in streamlit_app.py and 7× in med_suite.py. Anyone with repository access can read them. The chatbot even refers to the “Master Password”.	Move secrets to st.secrets or a secret manager, store only hashes, and use per-user authentication (SSO in Flux) with roles and an audit trail. Rotate both values now.

ID	Severity	Finding, location & evidence	Recommendation
S2	High	**Proprietary data could be exposed in the web port.** The product compositions (active % of PBTC, HEDP, ATMP, polymers) and k-factors are in plain code.	Keep them server-side and encrypted at rest. Never ship them in a client bundle or API response.
S3	High	**RO data and the RO model are not persisted.** load_database()/save_database() use Google Sheets only when target_file == LOCAL_DB_FILE (MED), lines 97 and 118. RO records and the RO RF/XGB model (joblib, line 1133) go to the ephemeral local disk, which Streamlit Cloud wipes on restart. When the model file is missing, the prediction becomes 0 (line 776), the residual % becomes 0, and the dashboard shows **“CLEAN”**, hiding fouling.	Persist RO data and models the same way as MED (or in the Flux database). If the model is missing, show “Model unavailable” instead of a status.
S4	High	**Silent write failures.** Bare `except: pass` around the Sheets connection and writes (init_db_connection, save_database; 8 bare excepts in streamlit_app.py) means “Record saved” is shown even when only the local file was written.	Report the storage target and any failure to the user, and log exceptions.
R1	High	**The RO “normalised twin” is not normalised.** Predicted permeate = f(Feed Flow, Feed TDS, Coagulant ppm, SMBS ppm) (lines 761–770). It has no temperature term (flux changes about 3%/°C), no feed pressure/NDP, no ΔP and no osmotic correction. Coagulant and SMBS have no physical link to permeate flow. With CIP thresholds at −4%/−5%, a few degrees of seasonal temperature change will trigger “Membrane CIP Required”.	Implement ASTM D4516 normalisation: normalised permeate flow, normalised salt passage and normalised ΔP, against a post-CIP baseline. Use the industry triggers (NPF −10–15%, ΔP +15%, NSP +5–10%). Keep ML only as a supplement.
R2	Medium	**Model accuracy is overstated.** R^2 is computed on the training data (lines 1092–1099), so RF/XGB will show close to 100%.	Use a time-based hold-out or cross-validation, and report MAE in m^3/h.
R3	Medium	**Bulk upload invents data.** Missing cells are filled with 450 m^3/h, 2000 ppm TDS, TSS 8, SDI 2.5, pH 7.2 and so on (lines 1175–1215). Pass/fail is then graded on these values, and a blank permeate flow becomes 0, which raises a false fouling alarm. The date is parsed with dayfirst=True (line 1182), unlike the ISO-first parsing fixed elsewhere.	Leave missing values as missing (NaN) and exclude them from grading and modelling. Reuse the shared ISO-first date parser.
R4	Medium	**Antiscalant is missing from the RO hub.** The RO Chemical Dosing tab covers only coagulant, flocculant and SMBS, and says that optimal dosing is “in development”. The chatbot says the same, although the Projection Engine exists.	Link the RO hub to the Projection Engine: pull the latest feed analysis, show the recommended and actual antiscalant dose, and log the dose.
R5	Low	**Smaller RO issues.** “Save Operational Record” appends duplicate dates (line 954). Remarks loaded from the database (ro_remarks) are not shown in the text box (ro_in_remarks). The chatbot keyword “ai” matches “explain”, “maintain” and “train” (line 476). The permeate pH guarantee of 7.0–7.5 is unusual for RO permeate (typically 5.5–6.5 before correction): confirm the specification.	Upsert by date, bind a single remarks key, match chatbot keywords on word boundaries, and confirm the guarantee table with the plant.
R6	Low	**MED chemical constants.** Antiscalant SG is 1.20 in MED (med_suite.py line 368) but 1.15 in the projection engine. AF_DENSITY = 0.02 is a dilution factor, not a density. LITRES_PER_LEVEL_UNIT = 23 is undocumented tank geometry.	Use a single product master table (SG, active %, price), and name the constants for what they are.

6. Build, deployment & code quality

ID	Severity	Finding, location & evidence	Recommendation
D1	High	**The desktop launcher exits immediately.** run_app.py starts Streamlit in a daemon thread, opens the browser, and the main thread then ends, which kills the server. Streamlit's CLI also registers signal handlers, which fail outside the main thread.	Run stcli.main() on the main thread, and open the browser from a threading.Timer.

ID	Severity	Finding, location & evidence	Recommendation
D2	High	**The executables are missing modules.** build-exe.yml bundles only streamlit_app.py. projection_engine.py, med_suite.py, calculator_tab.py, chembond_logo.png and .streamlit/config.toml are not added, so the build fails on import. plotly and kaleido are not installed in CI.	Add each file with --add-data (or use a .spec file), and add a smoke test that launches the built app in CI.
D3	Medium	**Dependencies.** requirements.txt has no version pins. “plotly.express” is not the plotly package (it resolves to the deprecated plotly-express shim). streamlit is not listed. fpdf is unused. oauth2client is deprecated (use google-auth / gspread.service_account_from_dict).	Pin versions (pip-tools or uv lock), list plotly and streamlit explicitly, and remove unused packages.
D4	Medium	**No tests or documentation.** README.md is empty, and there are no unit or regression tests for a calculation that drives customer proposals.	Add pytest regression tests for cases A–E plus PHREEQC cross-checks, and run them in CI. Write a README covering the method, assumptions, constants, sources and limitations.
D5	Low	**Structure.** projection_engine.py mixes UI and chemistry in one 900-line method, with functions copied three times. .devcontainer disables CORS and XSRF protection (acceptable only for local development).	Split it into a pure chemistry package (no Streamlit imports) that Flux can call through an API, plus a thin UI layer. That makes the web port and the tests share one engine.

7. Recommended order of work

Priority	Items
1 – before any further customer use	S1, S2 (secrets / IP), B3, B4 (report claims), B1, B2, M3 (false failures), M1 (3× sulfate error), B5
2 – model credibility	M11, M12 (antiscalant limits vs dose, calibrated with Chembond lab/field data), M2 (PHREEQC speciation), M4, M15
3 – plant monitoring	S3, S4 (persistence and silent failures), R1 (ASTM D4516 normalisation), R2, R3, R4
4 – engineering hygiene	D1–D5, M5–M10, M13, M14, M16, B6–B9, R5, R6

Validation suggestion: run 10–20 historical site analyses (with known scaling or no-scaling outcomes) through the corrected engine and through an established membrane projection program or PHREEQC, and publish the comparison internally before Flux presents the results as projections to customers.

Line numbers refer to commit 030a52a. The benchmark script used for the “verified” items is included in the repository next to this PDF (docs/review/benchmark_cases.py).